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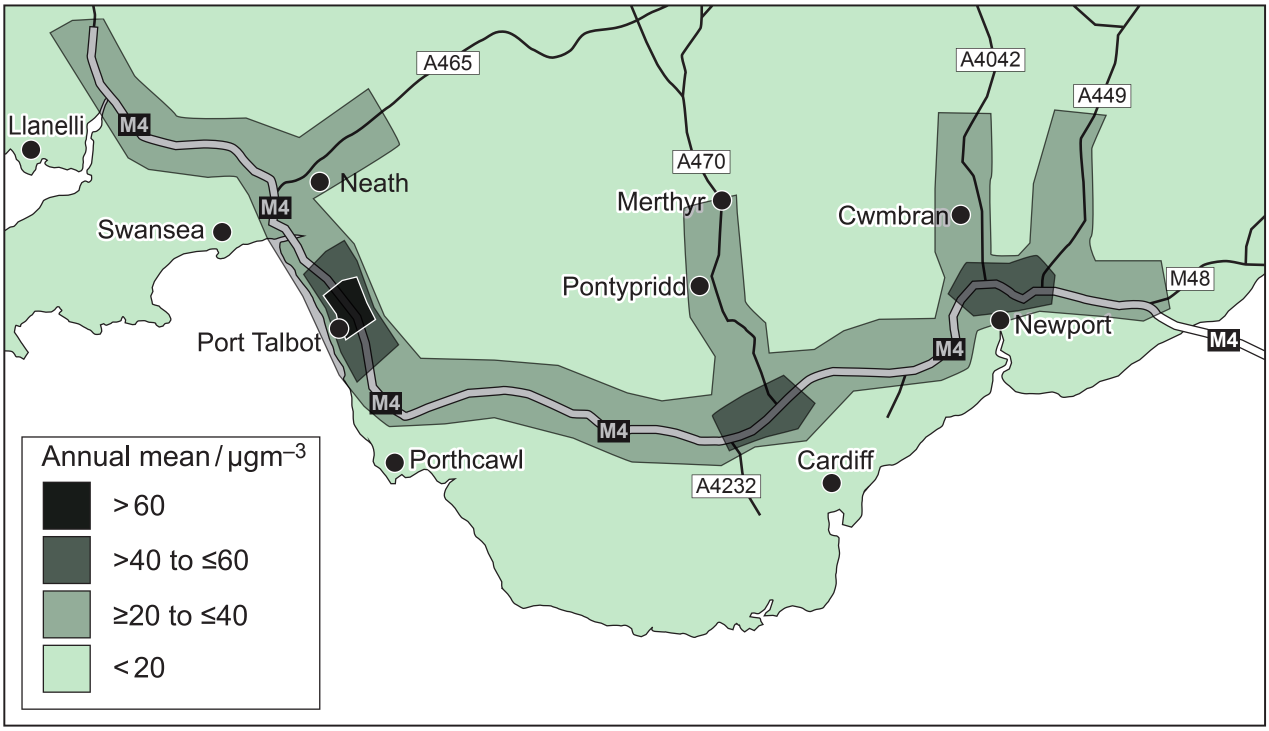
3. In Wales, over a thousand deaths are attributed annually to nitrogen dioxide (NO_2) pollution. Although the planetary boundary for chemical pollution remains to be determined, the World Health Organisation (WHO) suggests that the NO_2 annual mean value should not exceed 40 micrograms per cubic metre ($\mu g m^{-3}$).

The map in **Image 3.1** shows the background levels of airborne NO_2, measured by a network of air sampling machines located along the M4 motorway and major A-roads in South Wales. The key shows mean NO_2 levels in $\mu g\, m^{-3}$ for 2015.

Image 3.1



(a) (i) Describe how the distribution of measured NO_2 levels supports the hypothesis that motor vehicle exhausts are the main source of airborne NO_2. [1]

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(ii) I. **Draw a circle** on the map in **Image 3.1** to show the position on the M4 where traffic control measures were most urgently needed in 2015. [1]

II. Explain your choice in terms of WHO limits. [1]

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In Wales, Air Quality Management is the responsibility of local authorities. One local authority carried out an experiment to test the impact of closing a junction of the M4. They measured the concentration of NO_2 at 15 sites surrounding the junction for 4 months (April–July) before closing it and for 4 months after closing it (August–November).
A t-test value was calculated to assess whether the difference in mean NO_2 concentration was significant.
The results are summarised in **Table 3.2**.

Table 3.2

	April–July (before closure)	August–November (after closure)
Mean NO_2 concentration / $\mu g\,m^{-3}$	20.04	23.09
Number of measurements	15	15
Standard deviation	±4.14	±5.23
t-test value	1.769	
Degrees of freedom	28	

(b) The null hypothesis used in the experiment was that 'there was no significant difference in mean NO_2 concentration before and after closing the junction'.

Table 3.3

		Level of Probability						
		0.1	0.05	0.025	0.01	0.005	0.001	0.0005
Degrees of freedom	26	1.315	1.706	2.056	2.479	2.779	3.435	3.707
	27	1.314	1.703	2.052	2.473	2.771	3.421	3.690
	28	1.313	1.701	2.048	2.467	2.763	3.408	3.674
	29	1.311	1.699	2.045	2.462	2.756	3.396	3.659
	30	1.310	1.697	2.042	2.457	2.750	3.385	3.646

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(i) Use the information from **Table 3.2** and the probabilities shown in **Table 3.3** to decide whether to accept or reject the null hypothesis at a suitable probability level. Explain your answer. [4]

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(ii) State why the local authority might use the results of this experiment to justify keeping the motorway junction open. [2]

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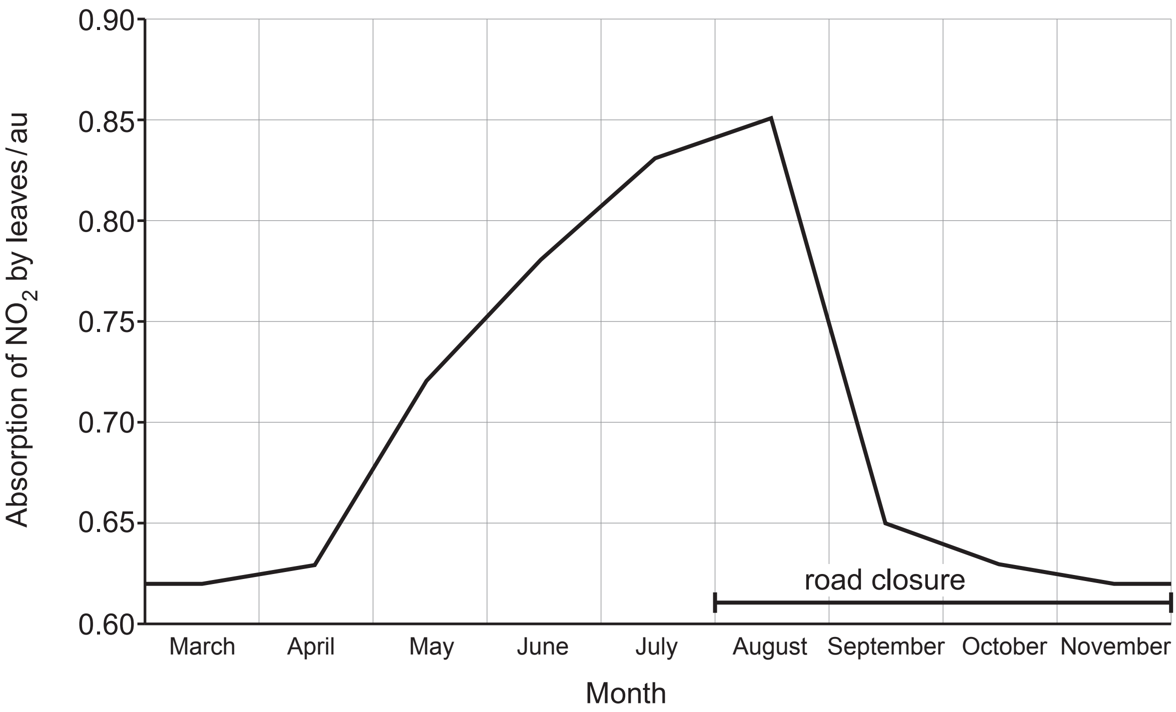
In a separate experiment, scientists measured the absorption of NO_2 by leaves on trees near the motorway at different times throughout the year. The trends are shown in **Graph 3.4**.

Graph 3.4



(iii) With reference to the results of the road closure experiment.

I. Explain how the **trends** shown in **Graph 3.4** affect confidence in the conclusion of the road closure experiment. [2]

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II. Explain how the design of the road closure experiment could be changed to take account of these findings. [1]

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NO_2 dissolves in rainwater to form nitrous acids which break down to form nitrite ions and wash into the soil.

(c) Name the soil bacteria that act on nitrite ions and explain how the soil bacteria change the nitrite ions so they are able to continue to circulate in the nitrogen cycle. [3]

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